

PHYSICS 525: Introduction to Plasmas

Section: 001

Plasma Exploration Project

Wave Interferometry

and

Electron Density of Plasma

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Abstract

This computational project develops a synthetic diagnostic model to simulate microwave interferometry measurements through a structured plasma. Discussion and theoretical background will touch on the importance of electron density measurements in fusion and plasma experiments along with a thorough examination of how we can find this using wave interferometry. The goal of this paper is to give insight into the many fields of physics that contribute to plasma physics.

Theoretical background

1. Electron Density in Plasmas

Measuring the electron density (n_e) is one of the most fundamental diagnostics in plasma physics and nuclear fusion experiments. Because macroscopic plasmas are essentially quasineutral (meaning the number of negative charges equals the number of positive charges, $n_e \approx n_i$), knowing the electron density inherently tells us the ion density. Knowing the electron density can tell us:

- a. Fusion Power Density: The power produced by a fusion plasma (e.g., Deuterium-Tritium mix) is proportional to the product of the interacting ion densities and the reaction cross-section.

$$P_{\text{fusion}} = n_D n_T \langle \sigma v \rangle E_f$$

Assuming a 50/50 Deuterium-Tritium mix and quasineutrality ($n_e \approx n_D + n_T$), the fusion power scales with the square of electron density: $P_{\text{fusion}} \propto n_e^2$. [1]

- b. The Lawson Criterion: To achieve ignition (plasma sustains its own temperature without external heating), the plasma must satisfy the Lawson criterion expressed as the product of density, temperature, and energy confinement time.

$$n_e T \tau_E \geq \text{constant}$$

Measuring n_e is required to know how close a reactor is to breakeven or ignition.

- c. Plasma Frequency (ω_{pe}): The natural at which electrons oscillate around the heavier, slower ions if they are displaced. This is purely a function of electron density,

$$\omega_{pe} = \sqrt{\frac{n_e e^2}{\epsilon_0 m_e}}$$

This frequency dictates how electromagnetic waves interact with the plasma. This is a critical tool that can be used to determine the electron density from wave interferometry.

- d. Debye Length (λ_D): the distance over which the mobility of the electrons shields out electric fields. It depends on both density and temperature,

$$\lambda_D = \sqrt{\frac{\epsilon_0 k_B T_e}{n_e e^2}}$$

A higher density means tighter, more effective shielding. For a gas to be formally considered a plasma, the physical size of the system must be much larger than the Debye length.

- e. Collision Frequency: The rate at which electrons collide with ions (ν_{ei}) scales roughly as,

$$\nu_{ei} \propto \frac{n_e}{T_e^{3/2}}$$

Density directly influences the collisionality regime of plasma. In high-density plasmas, transport is dominated by classic fluid-like diffusion. In low-density plasmas, trapped particles and complex kinetic turbulence dominate how heat escapes the core.

2. Wave Interferometry in Plasmas

The core principle behind plasma interferometry is that plasma acts as a dispersive medium. When an electromagnetic wave (like a microwave or a laser beam) passes through a plasma, it travels at a different phase velocity than it would in a vacuum. By measuring the resulting phase shift, we can calculate the electron density.

- a. Refractive Index of Plasma (N): For an electromagnetic wave with angular frequency ω traveling through an unmagnetized plasma, the refractive index is determined by the plasma frequency (ω_{pe}), the natural frequency at which free electrons oscillate in response to an electric field:

$$N = \sqrt{1 - \frac{\omega_{pe}^2}{\omega^2}}$$

Recall from earlier that the plasma frequency squared (ω_{pe}^2) is strictly proportional to the electron density (n_e):

$$\omega_{pe}^2 = \frac{n_e e^2}{\epsilon_0 m_e}$$

This relationship establishes two critical regimes:

- 1) Over-dense Plasma ($\omega < \omega_{pe}$): The refractive index becomes imaginary, meaning the wave cannot propagate and is reflected. The density at which $\omega = \omega_{pe}$ is called the critical density (n_c).
- 2) Under-dense Plasma ($\omega > \omega_{pe}$): The refractive index is real but less than 1. The wave propagates through the plasma, but its phase velocity ($v_p = c/N$) is faster than the speed of light in a vacuum. Interferometers must operate in this under-dense regime.

- b. Accumulating a Phase Shift: An interferometer splits a single coherent wave source into two identical beams.

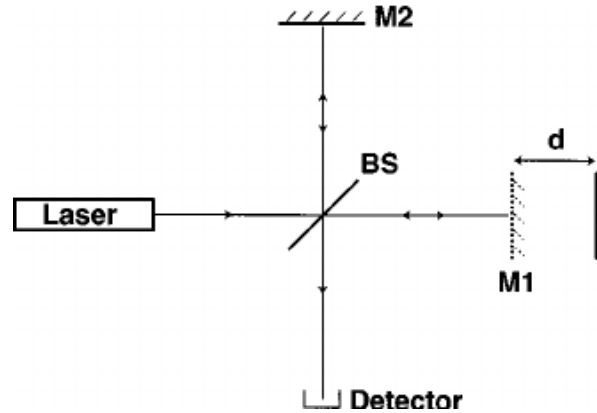


Figure 1.0 Michelson Interferometer [2]

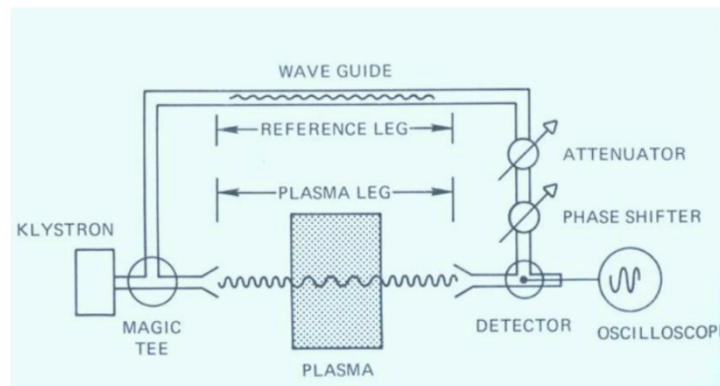


Figure 1.1 Mach-Zehnder Interferometer

- 1) Reference Beam: Travels through a vacuum (or air), isolated from the plasma.
- 2) Probe Beam: Travels a distance L directly through the plasma.

Because the plasma has a refractive index $N < 1$, the probe beam accumulates a different phase compared to the reference beam. The total phase (ϕ) of a wave traveling through a medium is integral of its wave number ($k = \omega N/c$) over the path length.

The measurable phase shift ($\Delta\phi$) between the two beams after they are recombined is:

$$\Delta\phi = \int_0^L k_{vac} dx - \int_0^L k_{plasma} dx$$

Substituting the wavenumbers:

$$\Delta\phi = \frac{\omega}{c} \int_0^L (1 - N(x)) dx$$

c. High Frequency Approximation

To extract the electron density mathematically straightforward, physicists usually choose a probe frequency that is much higher than the plasma frequency ($\omega \gg \omega_{pe}$). Under this assumption, we can use a Taylor series expansion on the refractive index:

$$N = \sqrt{1 - \frac{\omega_{pe}^2}{\omega^2}} \approx 1 - \frac{\omega_{pe}^2}{2\omega^2}$$

Substituting this approximation back into our phase shift equation:

$$\Delta\phi \approx \frac{\omega}{c} \int_0^L \left(1 - \left(1 - \frac{\omega_{pe}^2}{2\omega^2} \right) \right) dx = \frac{1}{2c\omega} \int_0^L \omega_{pe}^2 dx$$

d. Extracting the Electron Density

Finally, we substitute the fundamental definition of the plasma frequency back into the equation to reveal the direct relationship between the measured phase shift and the electron density:

$$\Delta\phi \approx \frac{e^2}{2c\epsilon_0 m_e \omega} \int_0^L n_e(x) dx$$

Because the charge of an electron (e), the speed of light (c), the vacuum permittivity (ϵ_0), the electron mass (m_e), and the frequency of our probe wave (ω) are all known constants, the phase shift is directly proportional to the integral of the electron density along the wave's path. This means a single interferometer beam does not give you the exact density at one specific point in space; rather, it provides the line-integrated electron density (often denoted as $\int n_e dl$). Since a single beam only gives total density along a chord, arrays of multiple interferometer beams passing through the plasma at different heights.

e. Abel's Inversion

The formula we have derived for the phase shift ($\Delta\phi \approx \frac{e^2}{2c\epsilon_0 m_e \omega} \int_0^L n_e(x) dx$), is known as a Volterra integral equation. Mathematician Niels Henrik Abel proved that an equation in the Volterra integral form, has a strictly defined analytical inverse. To extract the electron density n_e as a function of the phase shift, we need to perform what is known as Abel's inversion.

Let all physical constants be noted as the constant K : $\Delta\phi = K \int n_e(x) dx$

Assuming the plasma we are measuring is radially symmetric, the density n_e is a function of the radius r not x :

$$r = x^2 + y^2 \rightarrow x = \sqrt{r^2 - y^2}$$

$$dx = \frac{r}{\sqrt{r^2 - y^2}} dr$$

Substituting into the integral, we are now integrating from the closest point to the core out to the edge of the plasma where $r = a$. Because the wave from the interferometer passes through both the left and right halves of the plasma we can multiply the integral by 2:

$$\Delta\phi(y) = 2K \int_y^a \frac{n_e(r)r}{\sqrt{r^2 - y^2}} dr$$

Now we apply Abel's Inversion to obtain the electron density as a function of the phase change:

$$n_e(r) = -\frac{1}{\pi K} \int_r^a \frac{\frac{d(\Delta\phi)}{dy}}{\sqrt{y^2 - r^2}} dy$$

$$K = \frac{e^2}{2c\epsilon_0 m_e \omega}$$

Using this new function we now have the conversion we need to receive an output of the electron density from our wave interferometer measurements.

3. Simulation in PlasmaPy

To simulate this phenomenon, we first need to make a few assumptions. For our simulation we will be assuming that we are measuring an unmagnetized cold plasma that is collisionless. The profile of our electron density input will be a parabolic density where $n_e = n_0 \left(1 - \left(\frac{r}{a}\right)^2\right)$, with $a = 1$ m. This is done because the parabolic distribution is very distinct so that we may distinguish the relationships between the electron density, detection signal, phase shift, and compare the electron density input to our Abel reconstruction. We will define $n_0 = 5 \times 10^{18} \text{m}^{-3}$ and our interferometer probe frequency to be 100GHz.

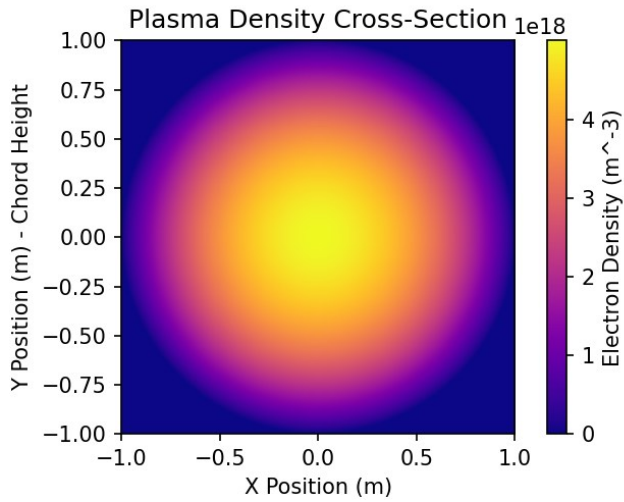


Figure 2.0 Input Electron Density

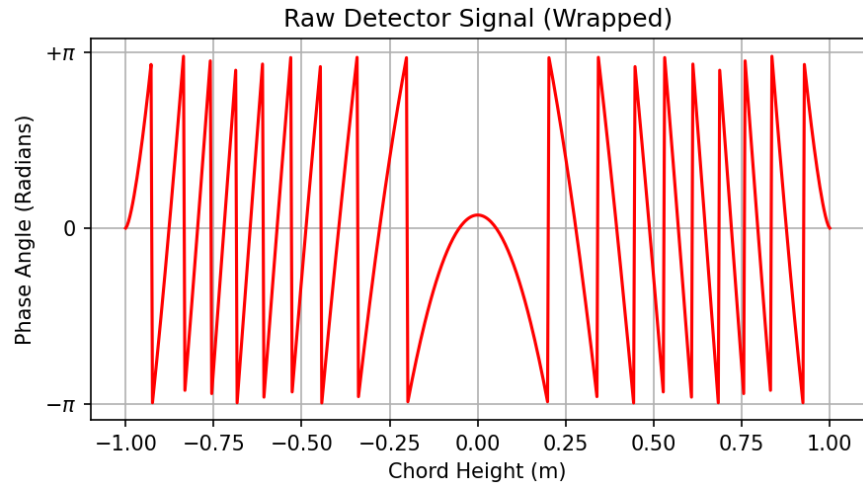


Figure 2.1 Interferometer Detector Signal

From the figure 2.0 we can see that the electron density is highest at the center (0,0). In figure 2.1 we can see that along the chord height (y-axis), we see fringes until the parabolic shape appears at $\pm 0.25\text{m}$. This is because the phase shift increases past π or 180° so it jumps down to $-\pi$ or -180° and keeps increasing until it completes another full period. To see its true parabolic nature, we can unwrap this signal and plot it. The process of unwrapping essentially uses sampling to capture each of the fringes, then reconstruction to connect the ends and create a single continuous curve

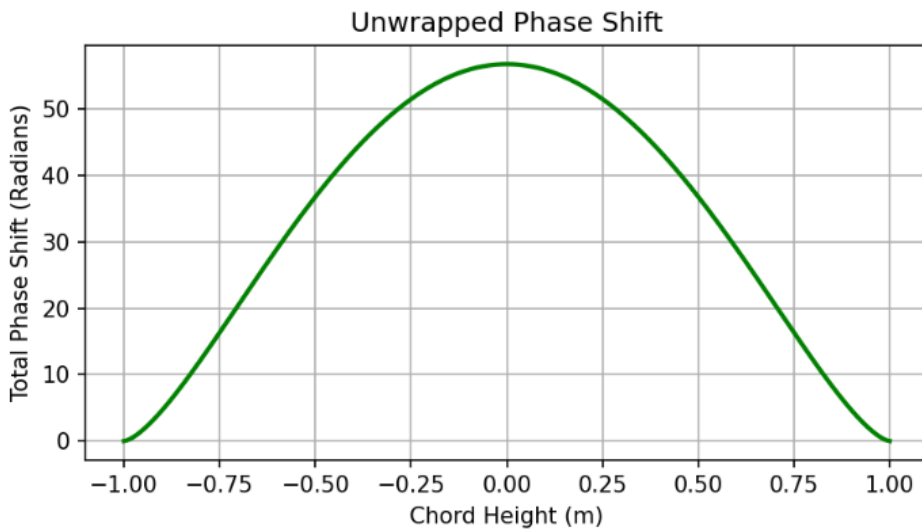


Figure 2.2 Unwrapped Phase Shift

As you can see in unwrapped detector signal in figure 2.2, the phase shift retains that parabolic shape which is consistent with our electron density. The stacked fringes from the detection signal are accumulated to get the total phase shift across the y-axis. Using this unwrapped signal of $\Delta\phi$ we can use Abel's inversion to obtain our measured curve for $n_e(r)$.

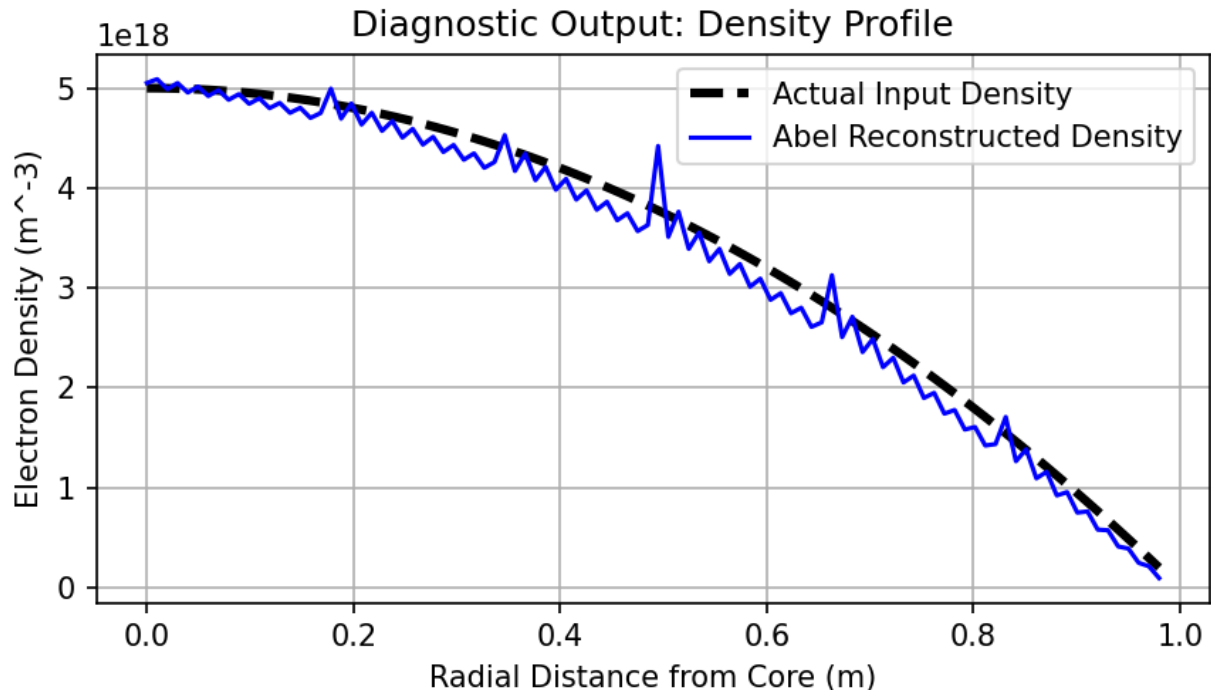


Figure 2.3 Electron Density Reconstruction

The horizontal axis is the radial distance from the center of the electron density profile. Taking advantage of the symmetry of this density profile allows us to reconstruct this plasma using single axis of chords. For a more realistic electron density profile you would need to use multiple axes of chords to reconstruct an accurate cross section measurement. Utilizing Abel's Inversion we can plot our radial electron density $n_e(r)$ as a function of the phase shift $\Delta\phi$ that was found using wave interferometry. As you can see in figure 2.3 the reconstructed signal is very similar to our input density with some noise created by the holes we needed to compensate for in the Volterra integral format used in Abel's inversion. This demonstration successfully shows how we can use wave interferometry to measure the electron density of plasma.

Conclusion

Through simulation we have examined the application of wave interferometry as a diagnostic technique for measuring electron densities in plasma environments.

Through the analysis of phase shifts experienced by electromagnetic waves propagating through a plasma medium, interferometry provides a non-invasive and highly accurate method for determining electron density profiles, a parameter central to understanding plasma behavior and performance.

The underlying principle, whereby the refractive index of a plasma is governed by the local electron density, allows measurable phase differences to be directly related to line-integrated electron density values. Wave interferometry has demonstrated its utility across a wide range of plasma conditions, from low-density laboratory discharges to the high-density, magnetically confined plasmas found in fusion devices such as tokamaks. Key advantages of interferometric diagnostics include their passive interaction with the plasma, their relatively straightforward calibration, and their capacity for real-time monitoring. However, limitations such as path-length ambiguity, sensitivity to vibrations, and the line-integrated nature of the measurement, which can obscure local density variations without additional tomographic reconstruction, must be carefully managed in experimental design.

Overall, wave interferometry remains one of the most reliable and widely deployed plasma diagnostic tools available. Its continued development, including advances in multi-chord configurations and the integration of phase-contrast imaging techniques, ensures that it will remain indispensable as plasma research pushes toward the demanding conditions required for sustained nuclear fusion. The technique exemplifies how classical wave physics can be harnessed to interrogate one of the most complex states of matter, yielding quantitative insight that is vital to both fundamental research and applied plasma engineering.

References

- [1] Popa, V. -A., et al. “EP-Stability-WF: An IMAS-Integrated Workflow for Energetic Particle Stability.” *arXiv.Org*, 14 June 2023, arxiv.org/abs/2306.08442.
- [2] Fox, Phillip & Scholten, R. & Walkiewicz, Mirek & Drullinger, R.. (1999). A reliable, compact, and low-cost Michelson wavemeter for laser wavelength measurement. *American Journal of Physics - AMER J PHYS*. 67. 10.1119/1.19335.
- (3) Gonçalves, Bruno. “Plasma Diagnostics.” *Sensors (Basel, Switzerland)* vol. 24,10 3257. 20 May. 2024, doi:10.3390/s24103257